Test Report

Toronto Island Ferry Analytics | executed 2026-10-18 13:13

1. Result

75 of 75 automated tests passed (0 failed, 0 errors, 0 skipped) in 19.6 s. pytest exit code 0; final line: *75 passed in 19.74s*. The Streamlit server started and answered health and page requests ('ok', HTTP 200, ready in 1.0 s, no traceback in its log).

2. Environment

Item	Value
Python	3.13.16
OS	Linux-6.18.44-fc-v139-x86_64-with-glibc2.39
Packages	streamlit 1.66.0, pandas 3.0.5, numpy 2.5.3, plotly 7.1.0, pytest 9.1.1
Command	python -m pytest (from ferry_analytics/)

3. Results by test module

Module	Tests	Passed	Covers
test_analytics	30	30	Aggregation, KPIs, rolling means, filters and boundaries, YoY, sales vs redemption, charts, CSV export
test_app_and_realdata	12	12	Real dataset facts, all dashboard pages headless, filters, error states, no-sales and minimal data
test_data_processing	33	33	Loading, schema, timestamps, duplicates, invalid values, gaps, DST, outliers, features

4. Verification checklist from the project brief

#	Check	Status	Evidence
1	Every imported module exists	Verified	Import audit below: all imports resolve to the standard library, a project module or a declared dependency
2	Every required dependency is declared	Verified	Same audit against requirements.txt (and requirements-reports.txt for report scripts)
3	Application functions connected correctly	Verified	Headless run of all 7 pages on the real dataset: no exceptions or error elements
4	App starts without syntax or import errors	Verified	streamlit run app.py: health 'ok', HTTP 200; AppTest start-up passes
5	Charts work with realistic and minimal data	Verified	test_all_charts_build_with_minimal_and_empty_data (full, 2-row and empty frames); tiny 3-row CSV through every page
6	CSV exports contain expected columns and filtered records	Partly verified	Export DataFrames tested for columns, row counts and totals. The browser download click itself was not automated
7	Graceful handling of missing/invalid input	Verified	Missing file, wrong extension, missing column, empty file and header-only file each produce a readable message
8	README commands match the structure	Verified	streamlit run app.py and python -m pytest were run from the documented directory; clean-environment install in section 6
9	No fabricated data in outputs	Verified	No synthetic data in the application path; synthetic data appear only inside tests; real-file totals are asserted in tests
10	Actual dataset processes successfully	Verified	261,538 rows; totals 12,785,293 redemptions and 12,972,051 sales asserted in tests

5. Import and dependency audit

File	Third-party / non-stdlib imports	Result
analytics.py	config, numpy, pandas	OK
app.py	analytics, config, data_processing, pandas, streamlit, visualizations	OK
config.py	-	OK
data_processing.py	config, numpy, pandas	OK
visualizations.py	config, numpy, pandas, plotly	OK
tests/conftest.py	data_processing, numpy, pandas, pytest	OK
tests/test_analytics.py	analytics, config, conftest, data_processing, numpy, pandas, pytest, visualizations	OK
tests/test_app_and_realdata.py	analytics, config, conftest, data_processing, pandas, pytest, streamlit	OK
tests/test_data_processing.py	config, conftest, data_processing, numpy, pandas, pytest	OK
reports/build_reports.py	PIL, metrics, playwright, reportlab	OK
reports/metrics.py	analytics, config, data_processing, matplotlib, numpy, pandas	OK

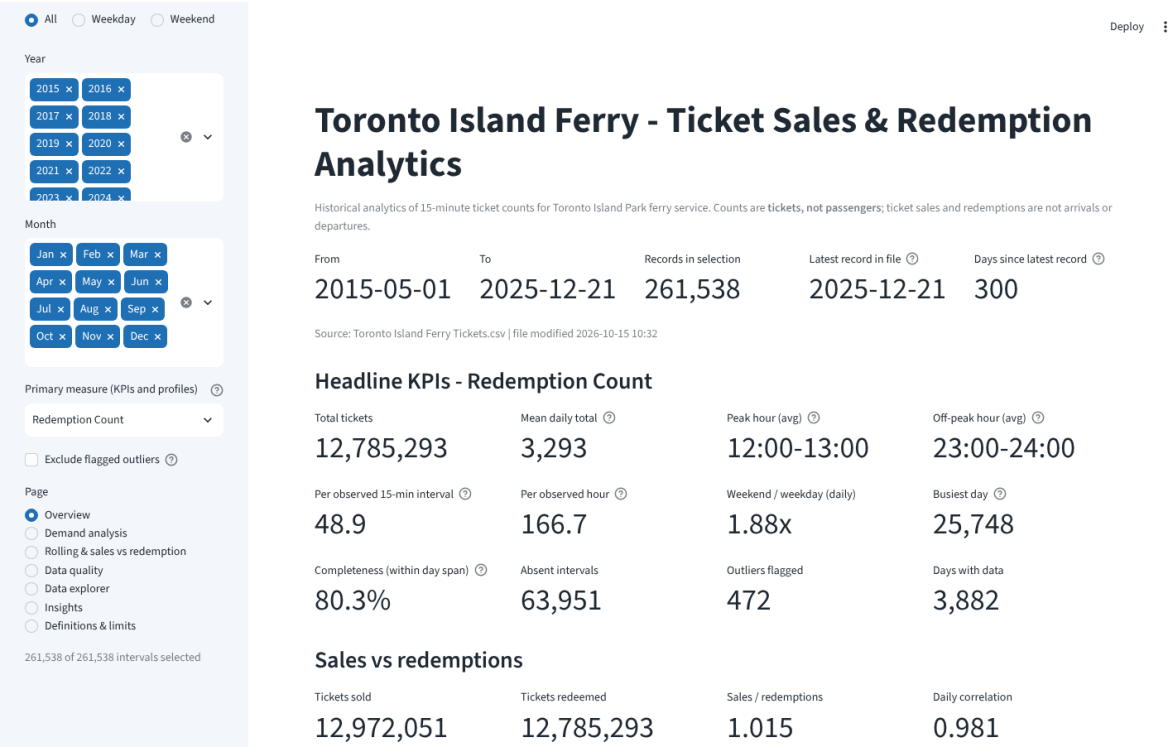
6. Clean-environment installation check

The project ZIP was extracted into an empty directory, a new virtual environment was created, only the declared dependencies were installed, and *python -m pytest* was run from the extracted *ferry_analytics* folder. This was done for the latest available versions and for the lowest versions declared in requirements.txt. The first minimum-version run exposed a real defect (the dashboard passed width="stretch" to st.dataframe on Streamlit 1.40, which only accepts pixel widths, so four app tests failed). It was fixed by detecting support from the function signature, and both environments then passed. The check was run on the code as packaged; the final ZIP differs only by the regenerated reports.

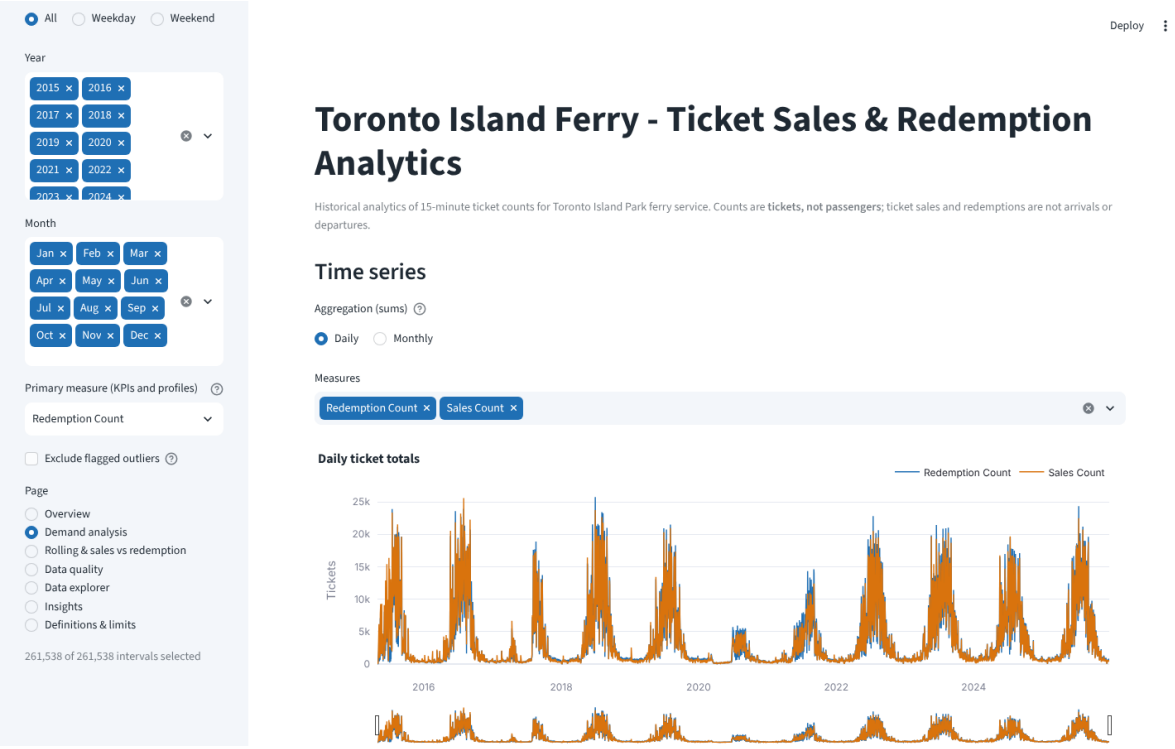
Environment	Python	Packages	pip install	Result
Latest versions (requirements.txt as written)	3.13.16	streamlit: 1.66.0, pandas: 3.0.6, numpy: 2.5.4, plotly: 7.1.0, pytest: 9.1.1	24 s	75 passed in 16.17s
Minimum declared versions	3.13.16	streamlit: 1.40.0, pandas: 2.2.3, numpy: 2.1.3, plotly: 5.24.1, pytest: 8.0.0	24 s	75 passed in 18.77s

7. Browser rendering check

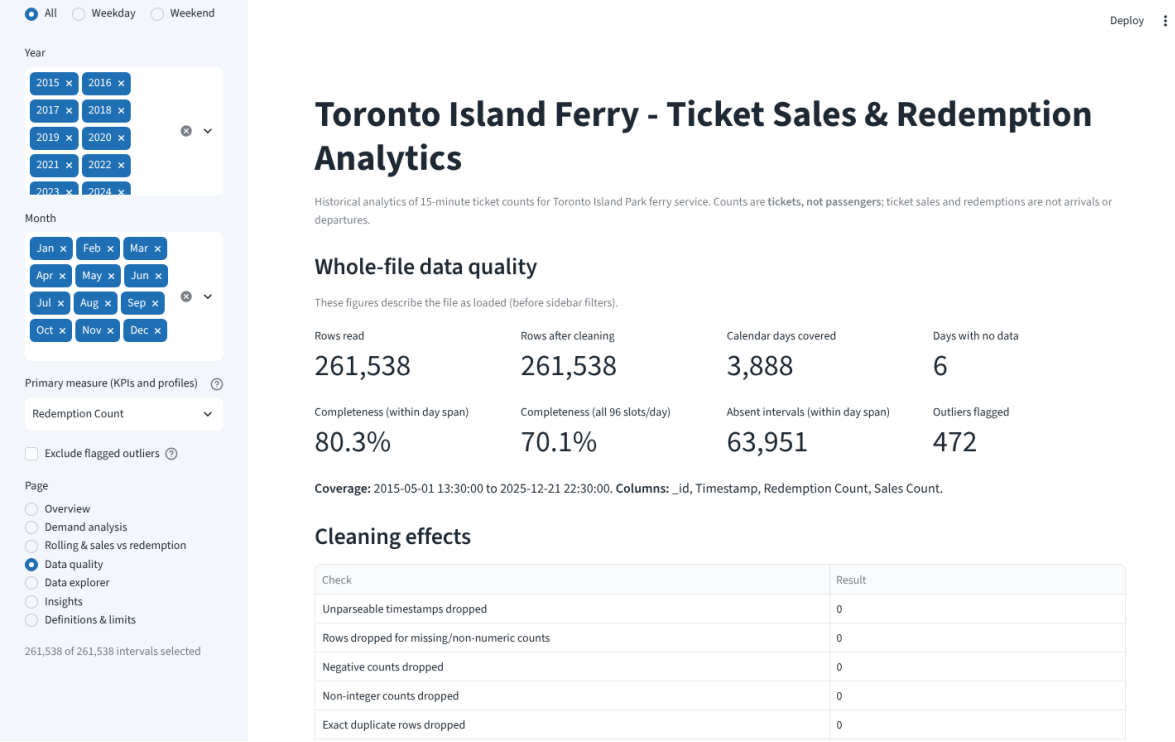
All seven pages were opened in headless Chromium against the live server on the real dataset; the expected heading rendered on each and the number of Streamlit exception boxes was 0. Three representative pages are shown; all seven screenshots are in reports/screenshots/.



Screenshot: Overview page (1440 x 900).



Screenshot: Demand analysis page (1440 x 900).



Screenshot: Data quality page (1440 x 900).

8. What was not verified

- Deployment to Streamlit Community Cloud was not performed from this environment; the README steps follow the platform's standard flow.

- Tests ran on Python 3.13.16 (Linux) only. The code uses no features newer than 3.11 but was not executed on 3.11, Windows or macOS. Versions between the tested minimum and latest were not each tested.
- The dashboard was checked visually in headless Chromium on a 1440 x 900 desktop viewport only; phone and tablet layouts were not checked.
- Browser file download and file-upload widget interactions were not automated.
- No load test beyond the 261,538-row dataset was performed.

9. Full list of test cases

Module	Test	Result	Time (s)
analytics	test_hourly_aggregation_sums_not_averages	passed	0.04
analytics	test_daily_and_monthly_aggregation	passed	0.03
analytics	test_aggregation_total_matches_source	passed	0.03
analytics	test_aggregate_bad_level_and_empty	passed	0.02
analytics	test_kpis_known_values	passed	0.04
analytics	test_offpeak_requires_enough_active_days	passed	0.04
analytics	test_kpis_empty	passed	0.00
analytics	test_kpis_missing_sales_measure	passed	0.02
analytics	test_seasonal_index_baseline_is_100	passed	0.03
analytics	test_low_to_peak_ratio	passed	0.04
analytics	test_peak_window	passed	0.03
analytics	test_peak_days_and_intervals	passed	0.04
analytics	test_sales_vs_redemption_values	passed	0.03
analytics	test_rolling_mean_values_and_edges	passed	0.03
analytics	test_rolling_default_windows_present	passed	0.04
analytics	test_filter_date_boundaries_inclusive	passed	0.04
analytics	test_filter_single_day_and_out_of_range	passed	0.04
analytics	test_filter_hours_inclusive	passed	0.04
analytics	test_filter_weekday_weekend	passed	0.03
analytics	test_filter_days_months_years	passed	0.04
analytics	test_filter_excludes_outliers	passed	0.03
analytics	test_filter_combination_consistent_with_kpis	passed	0.05
analytics	test_filters_on_empty_frame	passed	0.04
analytics	test_yoy_requires_full_months	passed	0.05
analytics	test_yoy_empty	passed	0.00
analytics	test_insights_contain_computed_numbers	passed	0.07
analytics	test_all_charts_build_with_minimal_and_empty_data	passed	1.67
analytics	test_sales_charts_show_unavailable_state_without_sales	passed	0.04
analytics	test_csv_export_columns_and_filtered_rows	passed	0.04
analytics	test_offpeak_threshold_is_90_percent_of_days	passed	0.08
app_and_realdata	test_real_dataset_loads_and_matches_known_facts	passed	1.82
app_and_realdata	test_real_dataset_aggregations_conserve_totals	passed	0.09
app_and_realdata	test_real_dataset_no_rows_in_spring_forward_hour	passed	0.00
app_and_realdata	test_app_starts_and_every_page_renders_on_real_data	passed	7.48
app_and_realdata	test_app_filters_update_kpis	passed	0.82
app_and_realdata	test_app_empty_filter_selection_handled	passed	0.72
app_and_realdata	test_app_missing_file_shows_friendly_error	passed	0.73
app_and_realdata	test_app_unsupported_file_shows_friendly_error	passed	0.65
app_and_realdata	test_app_missing_required_column_shows_friendly_error	passed	0.68
app_and_realdata	test_app_without_sales_column_degrades_gracefully	passed	1.61
app_and_realdata	test_app_minimal_dataset_all_pages	passed	1.58
app_and_realdata	test_default_path_is_relative_not_machine_specific	passed	0.00

Module	Test	Result	Time (s)
data_processing	test_load_from_path	passed	0.03
data_processing	test_load_from_filelike	passed	0.03
data_processing	test_missing_file	passed	0.00
data_processing	test_unsupported_format	passed	0.00
data_processing	test_empty_file	passed	0.00
data_processing	test_header_only	passed	0.00
data_processing	test_missing_required_column	passed	0.00
data_processing	test_missing_timestamp_column	passed	0.00
data_processing	test_timestamp_parsing_and_sorting	passed	0.02
data_processing	test_unparseable_timestamps_dropped_and_counted	passed	0.03
data_processing	test_tz_aware_timestamps_converted_to_toronto	passed	0.02
data_processing	test_exact_duplicates_dropped	passed	0.02
data_processing	test_repeated_timestamp_with_different_counts_is_summed_not_dropped	passed	0.03
data_processing	test_total_preserved_when_no_issues	passed	0.02
data_processing	test_negative_counts_dropped	passed	0.02
data_processing	test_missing_counts_dropped_not_zero_filled	passed	0.02
data_processing	test_non_integer_counts_dropped	passed	0.02
data_processing	test_all_invalid_raises	passed	0.01
data_processing	test_input_not_mutated	passed	0.02
data_processing	test_sales_column_optional	passed	0.02
data_processing	test_features	passed	0.02
data_processing	test_season_boundaries	passed	0.00
data_processing	test_off_grid_timestamps_floored_and_reported	passed	0.02
data_processing	test_dst_transition_days_detected	passed	0.00
data_processing	test_dst_flag_on_features	passed	0.02
data_processing	test_data_near_dst_does_not_crash	passed	0.02
data_processing	test_coverage_counts_absent_intervals	passed	0.03
data_processing	test_missing_calendar_days_listed	passed	0.02
data_processing	test_gap_table_finds_long_gap	passed	0.02
data_processing	test_regular_grid_zero_fill_and_nan	passed	0.03
data_processing	test_regular_grid_empty	passed	0.02
data_processing	test_outlier_flagged_but_retained	passed	0.04
data_processing	test_small_values_not_flagged	passed	0.03